

22-S2-Q1

Q(a) untrained	exercise
80	120

$$f(x) = 120x + (1-x)g(x)$$

$$g(x) = 50 + 30e^{-100x}$$

$$g(x) \rightarrow 80, x \text{ small}$$

$$g(x) \rightarrow 50, x \rightarrow 1$$

(i) Q: fill the Table with 20 value of x
optimum value for exercising

(ii) Q: fitness junction $f^*(x) \rightarrow 0$ to 1

(iii) Q: Difficulty of Find the optimum

Q(b) categorize

(i) share price $\rightarrow$ predict

(ii) odd size boxes $\rightarrow$ delivery

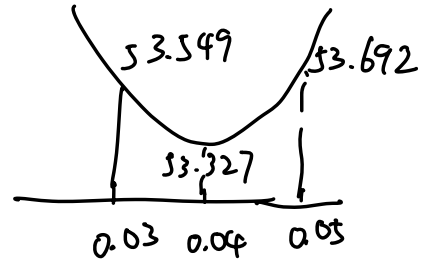
(iii) policy $\rightarrow$ making

(iv) Roster $\rightarrow$ security guards

Solution (i) ① minimize $f(x)$

$$f(x) = 120x + (1-x)g(x)$$

$$= 120x + (1-x)[50 + 30e^{-100x}]$$



②

x	g(x)	f(x)
0	80	80
0.01	61.0364	61.6266
0.02	54.0601	55.378
0.03	51.4936	53.549
0.04	50.5495	53.327
0.05	50.2021	53.6920
0.06	50.0743	54.2698
⋮	⋮	⋮
1	50	120

$$f(x) = 120x + 50 - 50x + 30e^{-100x} - 30xe^{-100x}$$

$$= 70x + 50 + 30e^{-100x} - 30xe^{-100x}$$

$$f'(x) = 70 - 3000e^{-100x} - 30[e^{-100x} - 100xe^{-100x}]$$

let $f'(x) = 0$ Using CASIO compute.

$$x = 0.0373, \quad f(x)_{\min} = 53.3039$$

about 3.7% of the day

Since there are $24 \times 60 = 1440$ minutes in a day, the optimum per day of exercise is

$$0.0373 \times 1440 = 53.712 \text{ mins/day}$$

(ii) We can use max-min normalize

$$f_{\max} = 120, \quad f_{\min} = 53.3$$

$$\text{So } f^*(x) = \frac{120 - f(x)}{120 - 53.3} = \frac{120 - f(x)}{66.7}$$

$$\text{when } f(x) = 53.3 \text{ (best)}, \quad f^*(x) = 1$$

$$\text{when } f(x) = 120 \text{ (worst)}, \quad f^*(x) = 0$$

x	$g(x)$	$f(x)$	$f^*(x)$
0	80	80	0.60
0.01	61.0364	61.6266	0.88
0.02	54.0601	55.378	0.97
0.03	51.4936	53.549	0.997
0.04	50.5495	53.327	0.999
0.05	50.2021	53.6920	0.994
0.06	50.0743	54.2698	0.9854
$\vdots$	$\vdots$	$\vdots$	
1	50	120	0

(iii) Solution

- ① It isn't a particularly difficult problem
- ② There is only one variable $x \in [0, 1]$
- ③ The function is smooth and unimodal
- ④ It can be solved by elementary numerical methods, such as bisection, Newton-Raphson, simple table search

(b) Question Type

(i) Free-optimisation problem (FOP)

Objective : $\min \text{MSE/MAE}$

Constraints : no mandatory rules

(ii) constraint and optimization problem (COP)

Objective : $\min$ wasted space

Constraints : no overlap, stay within truck

(iii) constraint and optimization problem (COP)

Objective : $\max$ economic growth, $\min$ unemployment rate

Constraints : fiscal-deficit ceiling, debt-to-GDP cap.

(iv) constraint and optimization problem (COP)

Objective : $\min$ imise wage cost, overtime

Constraints : shift-length limits, daily rest coverage per time slot